The MOND Acceleration Scale as the Cosmic Dynamical Acceleration

A scaling-MOND cosmology, its over-constrained web, and three honest bridges toward a theory of the dark sector

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Abstract. We examine the proposal that the MOND acceleration scale is the cosmic dynamical acceleration, $a_0 = (c/2)\sqrt{G\rho_c} = cH(z)/Z$ with $Z = 2\sqrt{8\pi/3} \approx 5.789$. Algebraically this is Milgrom's coincidence $a_0 \approx cH_0$ re-expressed through the Friedmann relation; its distinctive, coefficient-free consequence is that a_0 *evolves* as $a_0(z) = a_0(0)\cdot E(z)$ — an idea due to Milgrom (2014); our contribution is an explicit covariant realization and a proof of its CMB-safety. Fitting $a_0 \propto E(z)^p$ to the 2026 data (SPARC, Vărăşteanu, MUSE-DARK) gives $p = 0.80 \pm 0.17$; constant a_0 is excluded at 5σ in the nominal fit, but a jackknife and an inter-method systematic reduce this to ≈2σ, so we treat it as a suggestive hint, not a detection. From the AeST action, promoting a_0 to the aether expansion $a_0 = c\theta/3Z$, we show — by order-counting reinforced by the exact identity $\delta q^{00} = 0$ — that a_0 is absent from the linear perturbations, so running a_0 leaves the linear CMB exactly invariant (confirmed numerically: r_s = 144.3 Mpc, ℓ_A = 301.7). The boundaries are explicit: the O(1) coefficient is a posit; the Standard-Model constants do not follow; and the apparent-a_0 evolution is itself reproduced by ΛCDM simulations, so the amplitude does not discriminate. The one prediction that does — a redshift-weakening external field effect, $\eta = g_{ext}/a_0(z) \propto 1/E(z)$ — is forecast to require a dedicated $z > 4$ campaign.

1. The premise and its one novel framing

The empirical anchor is the MOND acceleration scale measured from 175 SPARC rotation curves, $a_0 = (1.1\text{–}1.2)\times10^{-10}$ m/s^2 [McGaugh et al. 2016]. The proposal writes it as half the surface gravity of the cosmic free-fall density:

$$a_0 = (c/2)\cdot\sqrt{G\,\rho_c} = c^2/(2R), \quad R = \sqrt{8\pi/3}\cdot c/H = c\cdot t_{ff}.$$

Via the Friedmann relation $H^2 = (8\pi/3)G\rho_c$, this equals $a_0 = cH/Z$ with $Z = 2\sqrt{8\pi/3}$. The $\sqrt{8\pi/3}$ factor is genuine Friedmann physics; the remaining factor of 2 — the Schwarzschild surface-gravity reading $c^2/2R$ — is the only un-derived element. The density / surface-gravity *form* appears to be novel (it is not the cH_0 or $\sqrt{\Lambda}$ form used in Milgrom's papers), but it is a re-expression of a known coincidence, not new physics, and the coefficient is not unique: the de-Sitter-horizon readings of Milgrom ($Z = 2\pi$) and Verlinde ($Z \approx 6$) fit the same data equally well.

2. The falsifiable prediction, confronted with data

Because a_0 tracks a cosmic density, it evolves. The prediction is coefficient-free — Z cancels in the ratio:

$$a_0(z) \,/\, a_0(0) = E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}.$$

Fitting $a_0(z) = A\cdot E(z)^p$ to the real compilation, marginalizing the normalization A, yields the result below. In the nominal fit, constant a_0 ($p = 0$) and the matter-only branch ($p = 3/2$) are each excluded at 5σ and the premise ($p = 1$) is consistent at 1.1σ. This significance is fragile, and we say so: a jackknife shows the rejection of constant a_0 rests almost entirely on the single $z \approx 0.9$ point (dropping it leaves 1.2σ), and the 1.7σ discrepancy between the two near-coincident local anchors (1.20 vs 1.69 at $z \approx 0$) evidences an inter-method systematic $\sigma_{sys} \approx 0.28$ that, folded in, reduces the rejection to ≈2σ — a suggestive hint, not a detection. Two honest caveats: (i) an evolving relation is also expected in ΛCDM — Magneticum simulations find apparent a_0 rising ×3 by $z = 2.3$, matching

$E(2.3) = 3.46$ with no fundamental evolution, so the data favor 'evolving over constant', not this framework over ΛCDM; (ii) de Graaff's $z \approx 6$ ratios $M_{dyn}/M_{\star} \sim 40$ are *not* support — the evolving-a_0 boost reaches only ~3 for such compact (near-Newtonian) systems.

z	a_0 (10^{-10} m/s²)	source
0.00	1.20 ± 0.26	SPARC — McGaugh, Lelli & Schombert 2016
0.05	1.69 ± 0.13	Vărăşteanu et al. 2025
0.90	2.38 ± 0.11	MUSE-DARK III 2026

Fit (a_0_decisive_pipeline.py): best $p = 0.80 \pm 0.17$; constant a_0 $\chi^2 = 27$ vs $E(z)$ $\chi^2 = 3.8$ vs $(1+z)$^1.5 $\chi^2 = 27.5$. A single clean a_0 at $z = 3$ (3%) sharpens p to $\approx 0.99 \pm 0.04$.

3. The over-constrained web

The content is a single dimensionless invariant, $a_0/cH = 1/Z = 0.173$, held fixed at every epoch and cast by dimensional analysis into six unit-costumes — an acceleration (the radial-acceleration relation), a length ($\ell = c^2/a_0 = Z \cdot R_H$), a surface density ($\Sigma = a_0/G$), a velocity (the baryonic Tully–Fisher zero-point), a temperature (T_{dS}/Z), and a time (the cosmic free-fall time). From the one premise about a set of relations follow as forced consequences. The web is *over-constrained*: the same a_0 must simultaneously fit the SPARC scale, the de-Sitter (Λ) floor $a_0(0)\sqrt{\Omega_\Lambda}$, and the intermediate-z points (Vărăşteanu, MUSE-DARK) — several determinations across cosmic time keyed to one number (the relation $H_0 = Za_0/c$ is an identity, not an independent check). We use this to draw the honest line: a real relation is z-invariant (a_0/cH stays $1/Z$); a coincidence between cosmic numbers is not. The same test that keeps these edges *rejects* the Standard-Model-constant 'derivations' (Section 5).

4. Three bridges toward a theory of the dark sector

Bridge 1 (relativistic completion — CMB-safety, established). The covariant home is the Aether–Scalar–Tensor theory [Skordis & Złošnik 2021], the one relativistic MOND that fits the CMB. Promoting its fixed a_0 to $a_0(z)$ by coupling to the aether expansion $\theta = \nabla \cdot A = 3H$ is a minimal modification. From the AeST action, a_0 enters only the spatial free-function term $F \propto Y^{3/2}/a_0$; on the FRW background the scalar is temporal, so $Y = O(\delta\varphi^2)$ and the a_0-term is $O(\delta\varphi^3)$ — absent from the linear equations. The one term the standard order-counting omits, $\delta q^{00} \cdot (d\varphi/dt)^2$, also vanishes exactly: the unit-timelike constraint forces $\delta q^{00} = \delta g^{00} + 2A^0\delta A^0 = 2\Psi - 2\Psi = 0$, even with the aether perturbed. Hence running a_0 leaves the linear CMB and matter power spectrum *exactly* invariant. A direct solution of the linear Einstein–Boltzmann system confirms this numerically (the running-a_0 effect is 0) and reproduces the Planck sound horizon $r_s = 144.3$ Mpc and acoustic scale $\ell_A = 301.7$. The CMB-fitting dust mode lives in the separate temporal sector $K(Q)$ and is a_0-independent.

Bridge 2 (the coefficient — reported straight, does not close). Horizon thermodynamics derives the evolution $a_0 \propto H$ route-independently and the order of magnitude, but does *not* pin the O(1) coefficient: legitimate matchings give $Z = 1$, 2π, ≈ 6, and 5.789, clustering within 8.5% — inside a_0's ~20% systematic. The framework's 'Schwarzschild' reading is not self-consistent (the enclosed mass is $8\pi/3$ times the Schwarzschild mass for R, so R is not a real horizon). The factor of 2 remains a posit. This costs the framework nothing, because the falsifiable prediction is the *evolution*, which is Z-independent.

Bridge 3 (dark matter = dark energy — sourced, falsifiable). In AeST the single temporal function $K(Q) = -2\Lambda + K_2(Q-Q_0)^2$ carries both the dark-matter-mimicking dust mode ($\rho \propto a^{-3}$) and dark energy (-2Λ) — one function, two terms, tied by the evolving a_0 that floors at the Λ value. The test is a_0-cosmography: $H(z) = Z \cdot a_0(z)/c$ reconstructs the expansion from galaxy dynamics. It lands on the Planck/local band at $z = 0$ and shows the known ~25% tension at $z \approx 0.9$.

5. Boundaries — what is explicitly not claimed

This is a candidate theory of the *dark sector* (gravity, dark matter, dark energy), not a Theory of Everything. The Standard-Model constants do not connect: a 34,000-formula search hits an arbitrary O(100) target to 0.004% about 20% of the time, 52 of 64 published Z^2 'derivations' contain no Z at all, and a *random* number reproduces α^{-1} and the mass ratios to the same precision as 32π/3. Those retrodictions carry ≈ 0 bits and are not part of this framework. Likewise the coefficient is a posit, and the full CMB peak heights require a dedicated Boltzmann implementation. Stating these limits is what keeps the surviving result trustworthy.

6. The decisive test

Pinning the exponent p by measuring a_0 at $z > 2$ tests evolution against constant a_0, but does *not* separate this framework from ΛCDM, whose apparent a_0 also rises as E(z) (Section 2). The deep-MOND cascade ($M_{dyn}/M_{\star} \propto \sqrt{E}$, $\sigma \propto E^{1/4}$, Tully–Fisher zero-point ∝ −log E) shares this degeneracy. The one discriminating observable is the external field effect: with $a_0 \to a_0(z)$, $\eta = g_{ext}/a_0(z) \propto 1/E(z)$, so the EFE *weakens* with redshift — distinct from ΛCDM (no host effect) and constant-a_0 MOND (constant EFE). Solving the MOND EFE, the distinctive signal is ≈0.1 dex, below the per-galaxy scatter (~0.3 dex); a 3σ detection needs ~600–1600 extended, environment-classified, kinematically-resolved galaxies at $z > 4$ — a dedicated multi-cycle JWST/ALMA(/ELT) program. That measurement, not a single pointing, decides it.

References

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Reproducibility: real_research/ — a0_powerlaw_confrontation.py, bridge1_linear_boltzmann.py, reviews/theta_3H_coupling.py ($a_0 = c\theta/3Z$, $\delta q^{00}=0$), reviews/stresstest_piece3_evolution.py (5σ→2σ), reviews/efe_evolution_forecast.py, reviews/parameter_space_map.py, false_discovery_rate.py; full assessment in COMPLETE_ASSESSMENT.md (github.com/carlzimmerman/zimmerman-formula).